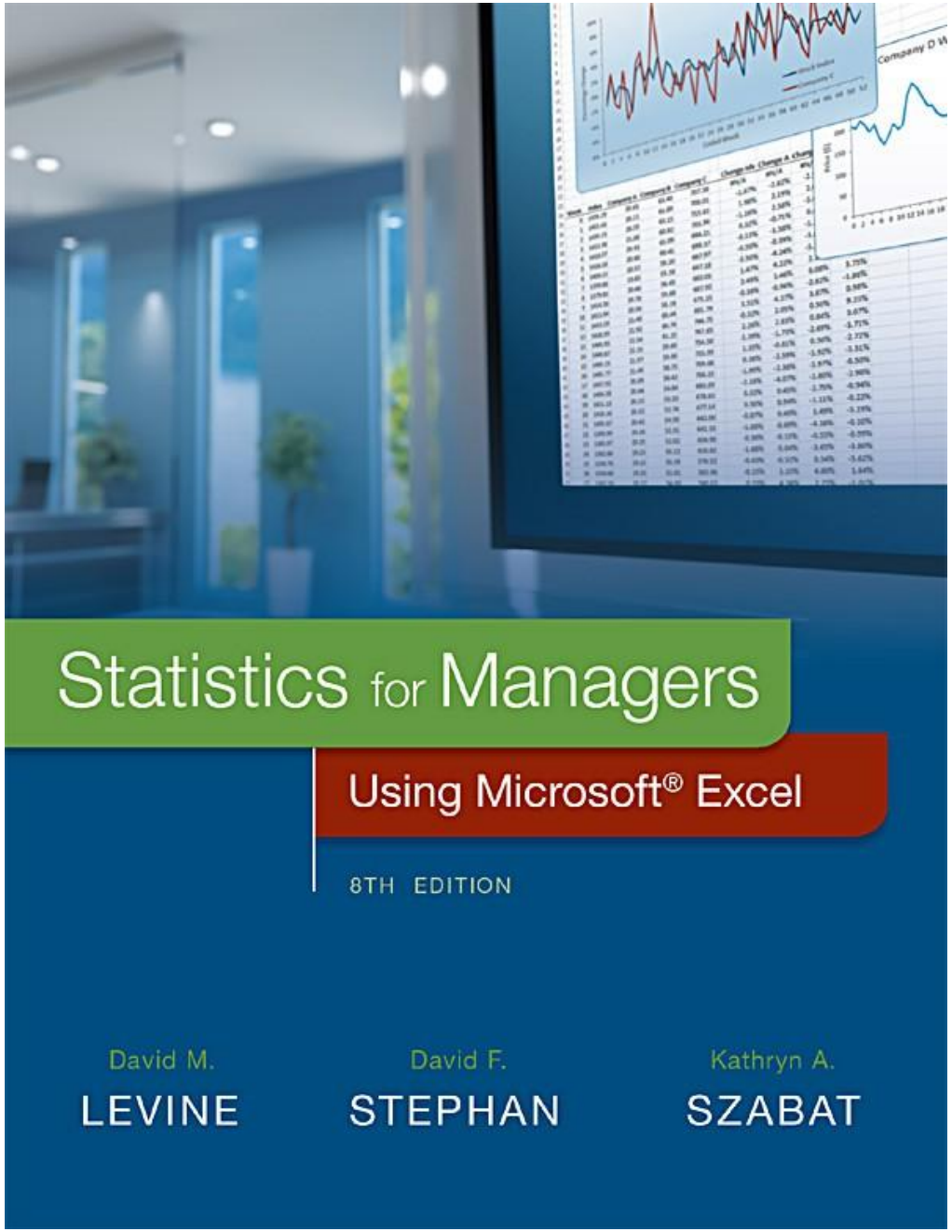


CHAPTER 13

Simple Linear Regression

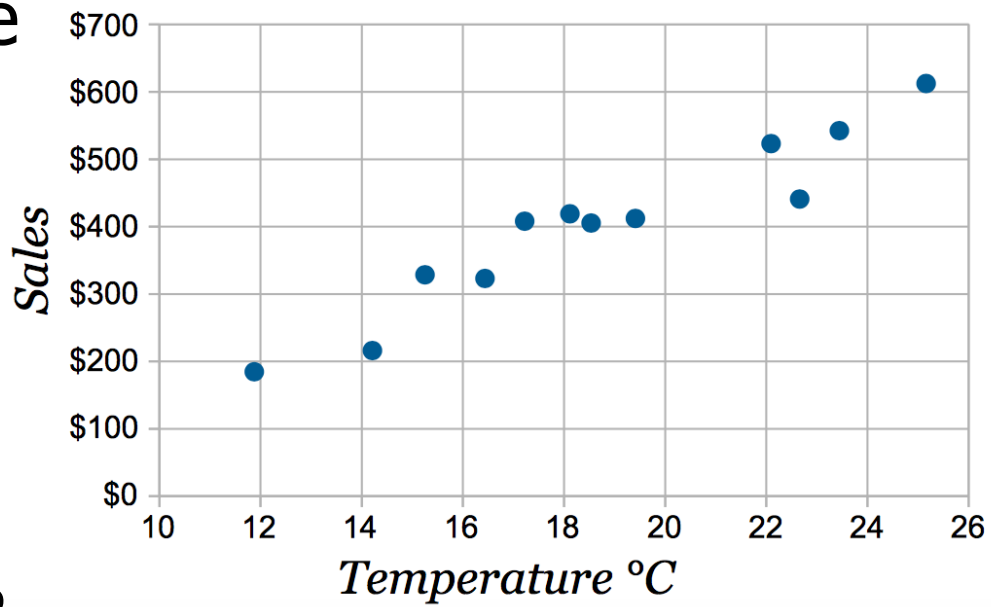
PAUL ANDREW GORGONIO

Correlation vs. Regression

A scatter plot can be used to show the relationship between two variables.

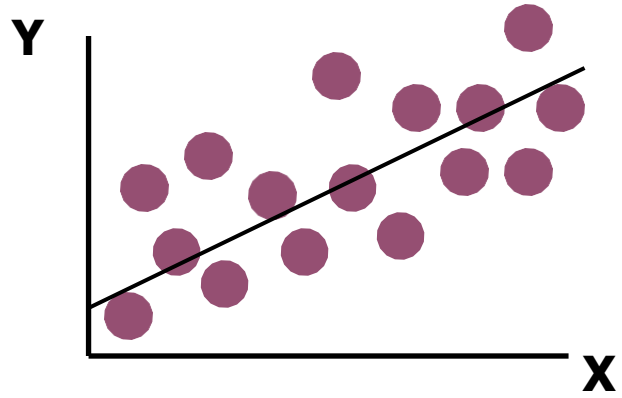
Correlation analysis is used to measure the strength of the association (linear relationship) between two variables.

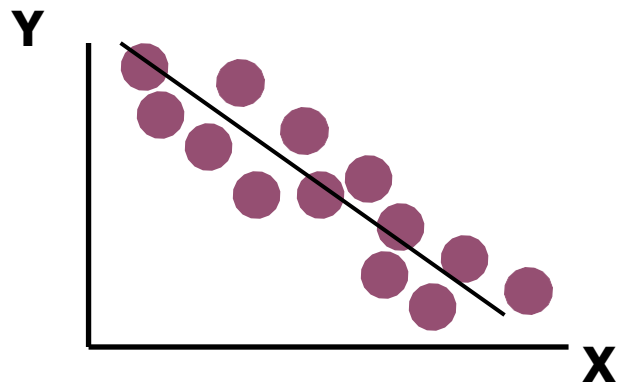
- Correlation is only concerned with strength of the relationship.
- No causal effect is implied with correlation.
- Scatter plots were first presented in Ch. 2.
- Correlation was first presented in Ch. 3.



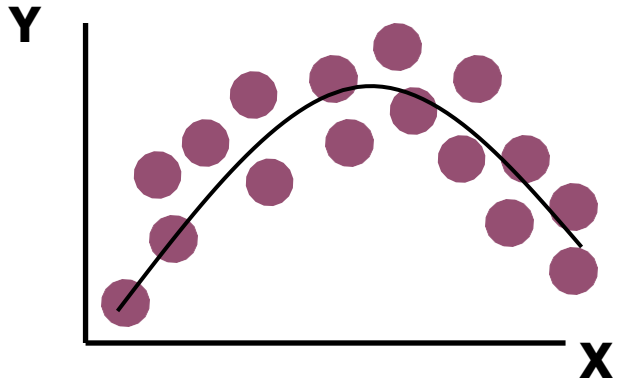
Types of Relationships

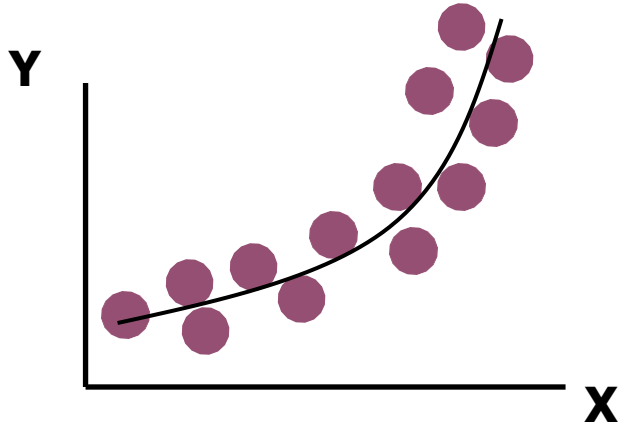
Linear relationships





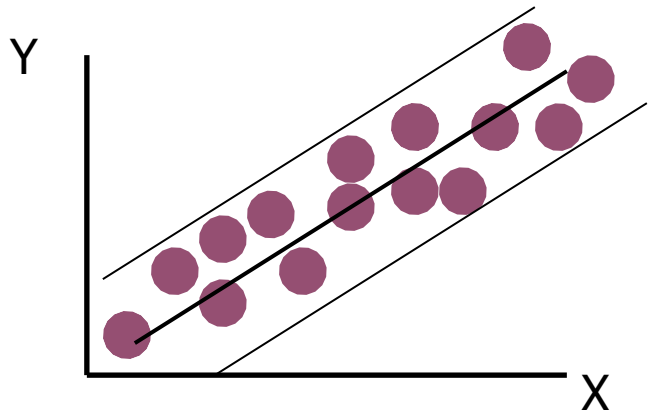
Curvilinear relationships

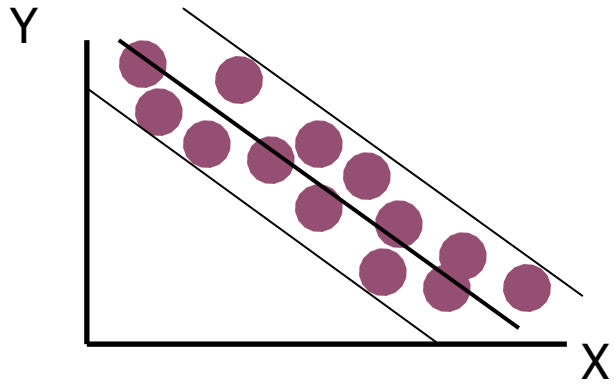




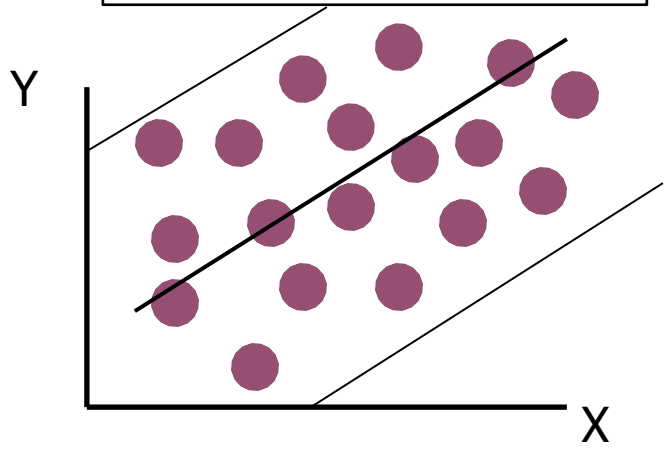
Types of Relationships

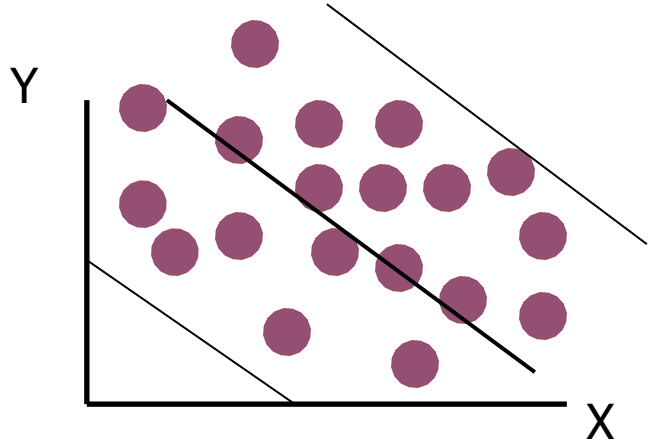
Strong relationships



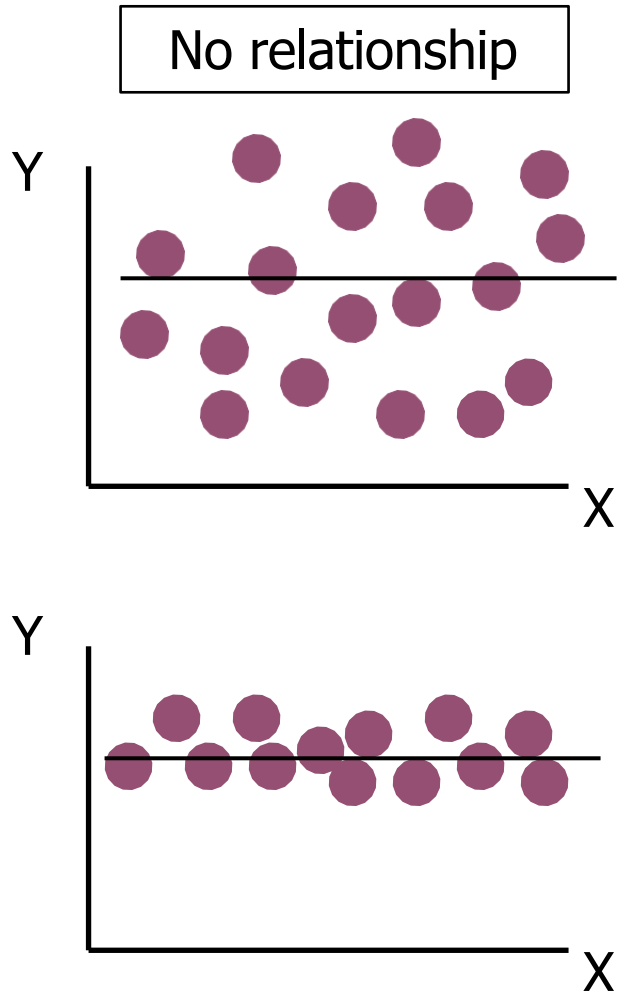


Weak relationships





Types of Relationships



Introduction to Regression Analysis

- Regression analysis is used to:
  - Predict the value of a dependent variable based on the value of at least one independent variable.
  - Explain the impact of changes in an independent variable on the dependent variable.

Dependent variable: the variable we wish to predict or explain.

Independent variable: the variable used to predict or explain the dependent variable.

Independent and Dependent Variables

Dependent variable:

the variable we wish to predict or explain

Independent variable:

the variable used to predict or explain the dependent variable

Examples:

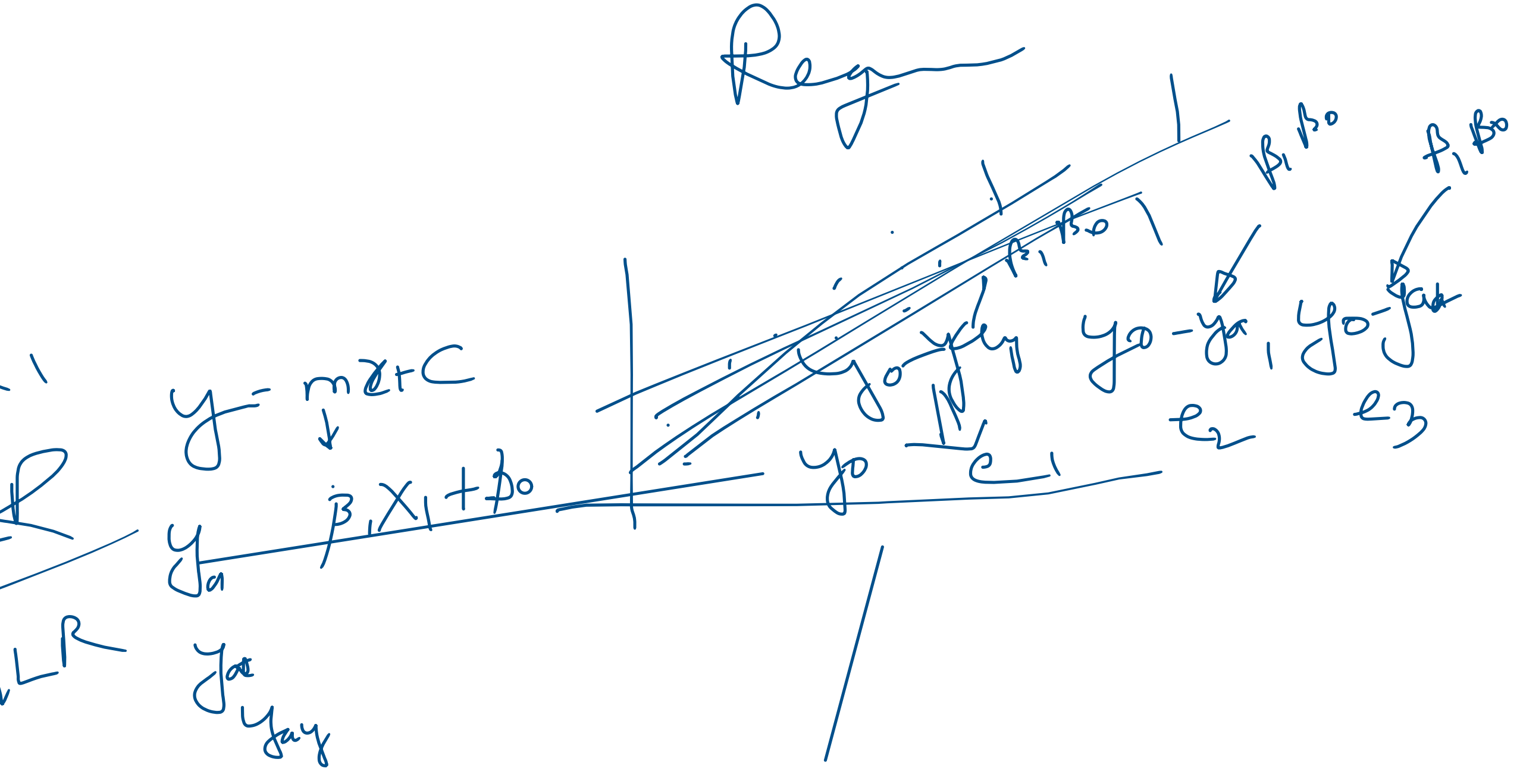
1. Influence of **different quantities of nutrient intake** on the **growth of an infant**
2. **Number of profiled customers** who live no more than 30 minutes from a store and **store's annual sales**

Simple Linear Regression Model

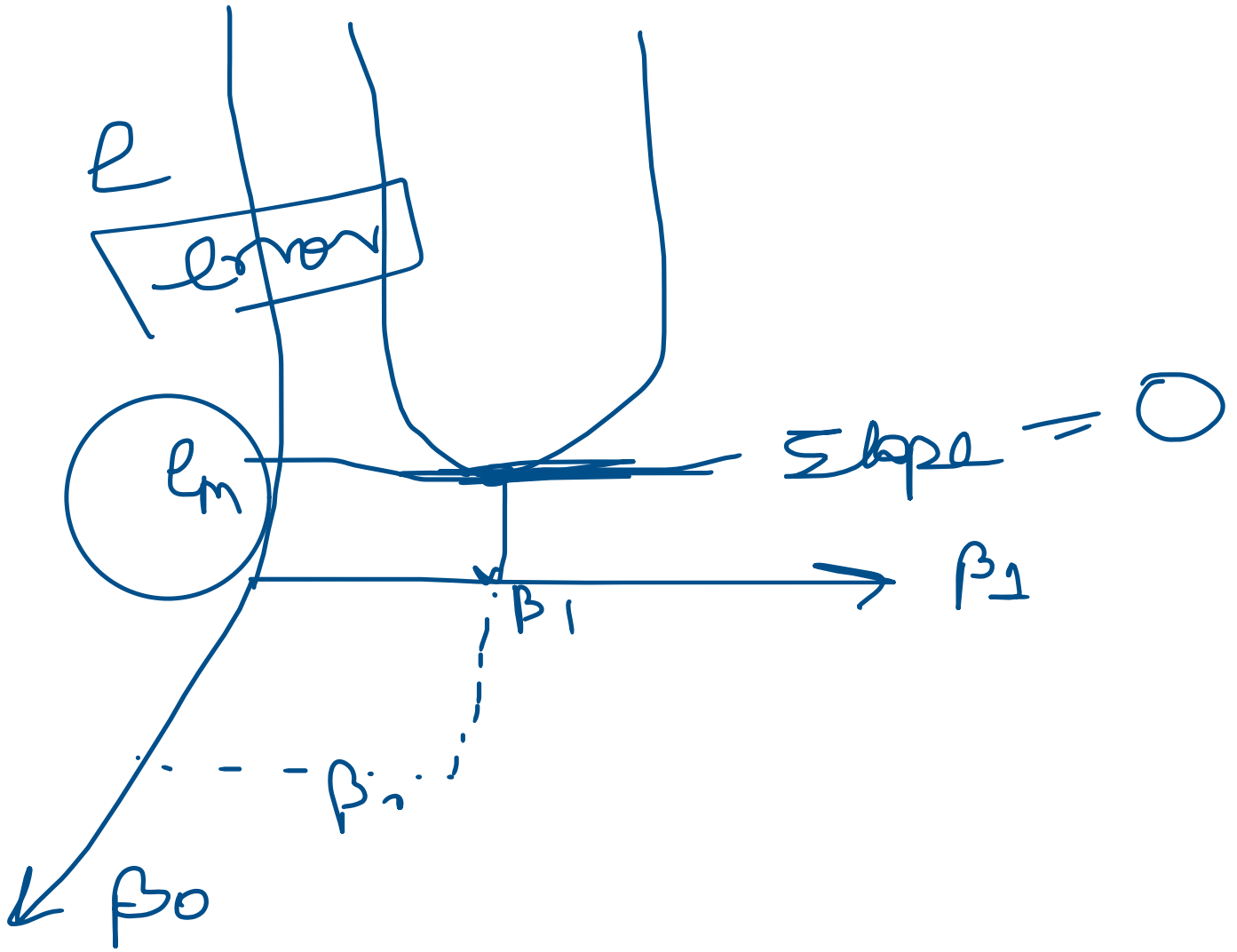
Only one independent variable, X.

Relationship between X and Y is described by a linear function.

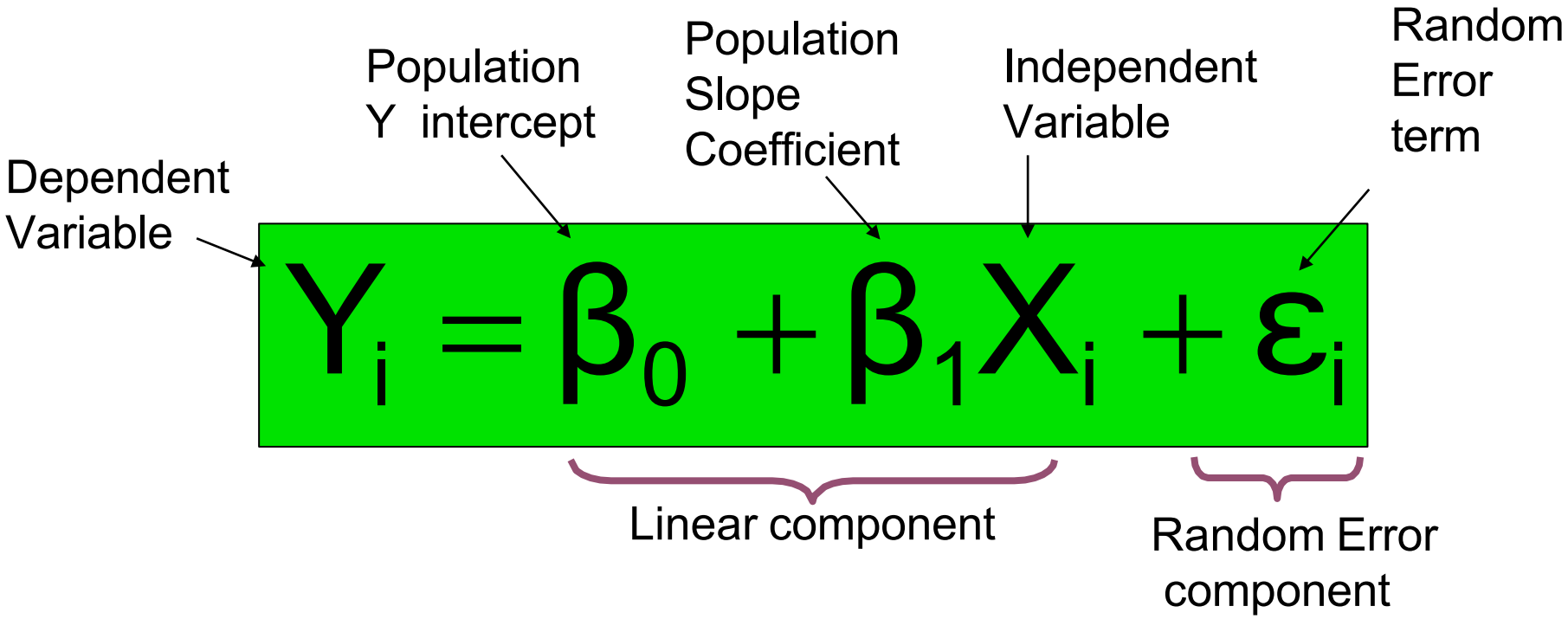
Changes in Y are assumed to be related to changes in X.


Reg
y = mx+C
β1X1+β0
y0
e1
e2
e3
β1β0
β1β0
β1β0
y0-ya
y0-ya
y0-ya
LR
ya
yay

$\hat{y} = \beta_1 x_1 + \beta_0$

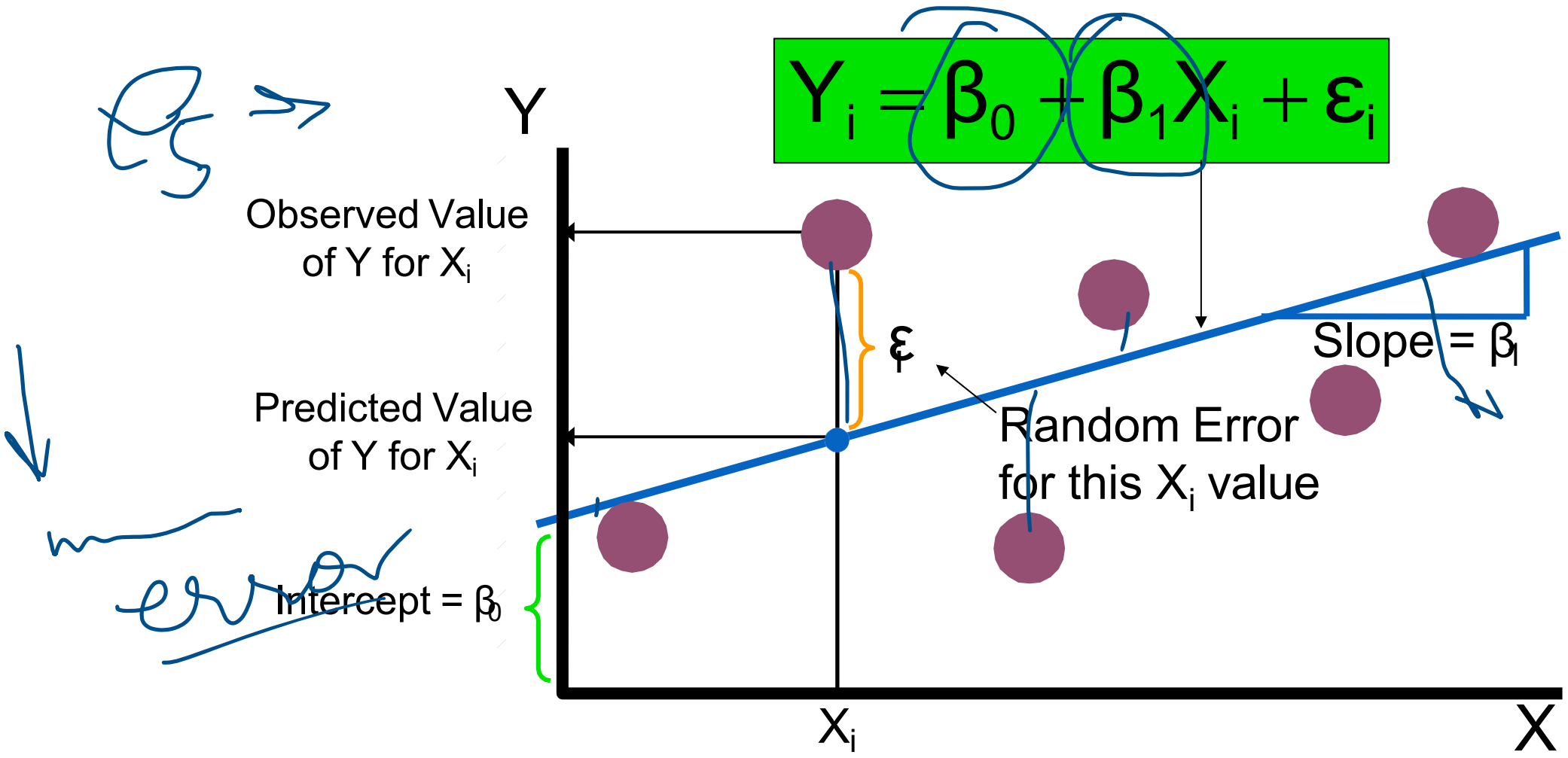


Simple Linear Regression Model



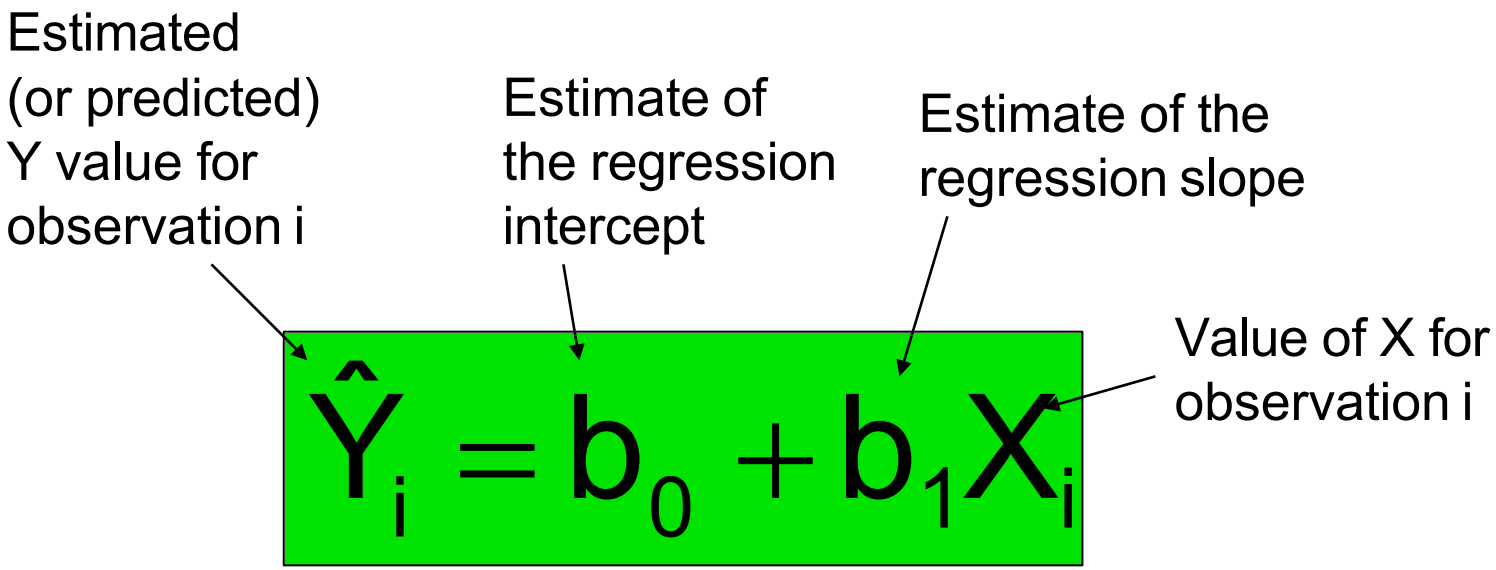
Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



Simple Linear Regression Model: Prediction Line

The simple linear regression equation provides an estimate of the population regression line.



The Least Squares Method

- b_0 and b_1 are obtained by finding the values that minimize the sum of the squared differences between Y and $\hat{Y}$:

$$\min \sum (Y_i - \hat{Y}_i)^2 = \min \sum (Y_i - (b_0 + b_1 X_i))^2$$

- The coefficients b_0 and b_1, and other regression results in this chapter, will be found using Excel.
- The calculations for b_0 and b_1 are not shown here but are available in the text.

Interpretation of the Slope and the Intercept

- b_0 is the estimated mean value of Y when the value of X is zero.

- b_1 is the estimated change in the mean value of Y as a result of a one-unit increase in X.

Simple Linear Regression Example

- A real estate agent wishes to examine the relationship between the selling price of a home and its size (measured in square feet).

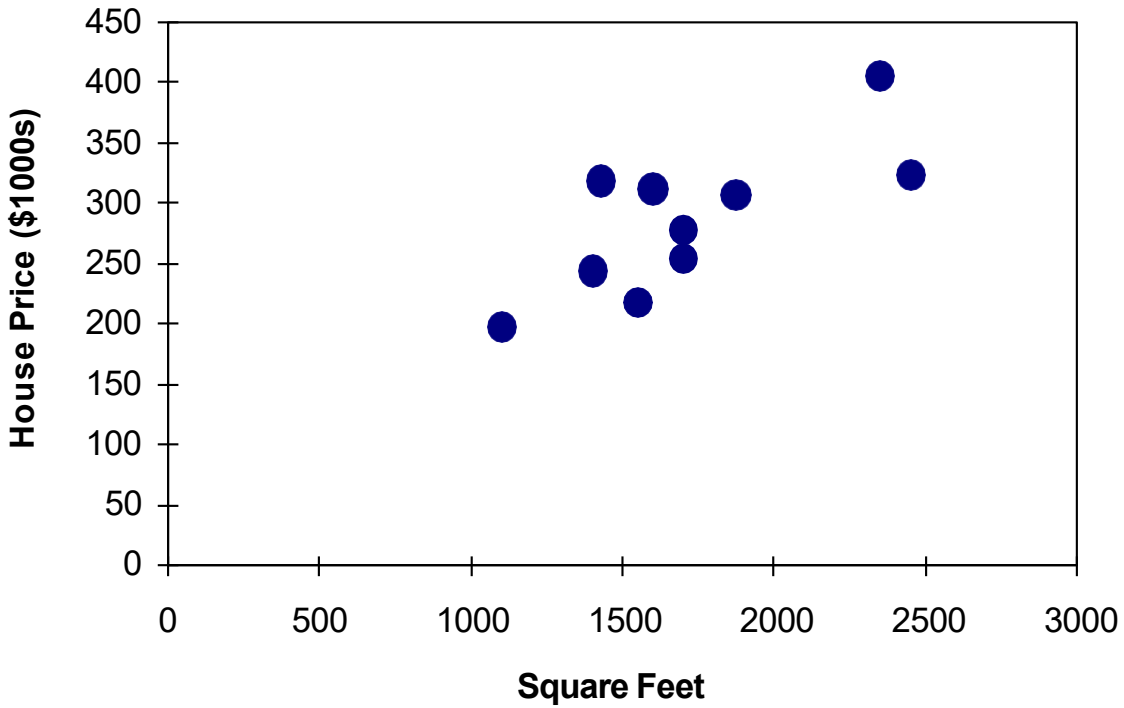
- A random sample of 10 houses is selected.
  Dependent variable (Y) = house price in $1,000s.
  Independent variable (X) = square feet.

Simple Linear Regression Example: Data

House Price in $1000s (Y)	Square Feet (X)
245	1,400
312	1,600
279	1,700
308	1,875
199	1,100
219	1,550
405	2,350
324	2,450
319	1,425
255	1,700

Simple Linear Regression
Example: Scatter Plot

House price model: Scatter Plot



Simple Linear Regression
Example: Excel Output

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

The regression equation is:

$$\widehat{\text{house price}} = 98.24833 + 0.10977\ (\text{square feet})$$

ANOVA

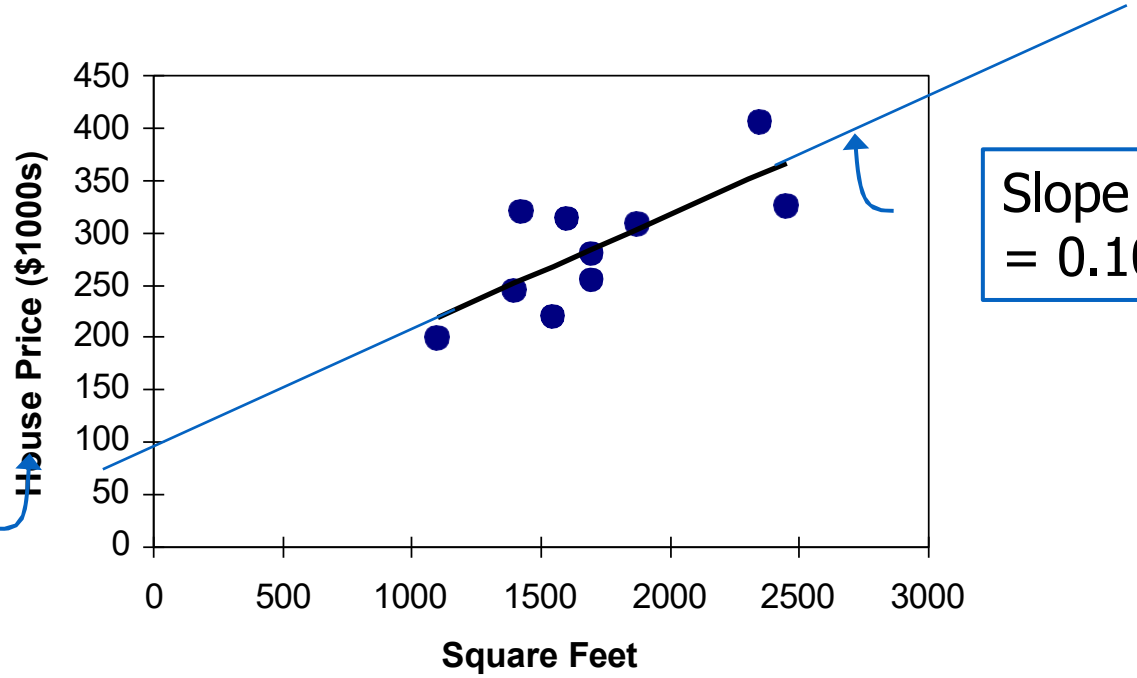
	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Simple Linear Regression
Example: Graphical Representation

House price model: Scatter Plot and Prediction Line





Intercept = 98.248

Slope = 0.10977

$$\widehat{\text{house price}} = 98.24833 + 0.10977\ (\text{square feet})$$

Simple Linear Regression
Example: Interpretation of b_o

$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

- b_0 is the estimated mean value of Y when the value of X is zero (if X = 0 is in the range of observed X values)
- Because a house cannot have a square footage of 0, b_0 has no practical application

Simple Linear Regression Example: Interpretation of b_1

$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

- b_1 estimates the change in the mean value of Y as a result of a one-unit increase in X.
- Here, b_1 = 0.10977 tells us that the mean value of a house increases by 0.10977($1,000) = $109.77, on average, for each additional one square foot of size.

Simple Linear Regression
Example: Making Predictions

Predict the price for a house with 2,000 square feet:

$$\widehat{\text{house price}} = 98.25 + 0.1098\,(\text{sq.ft.})$$

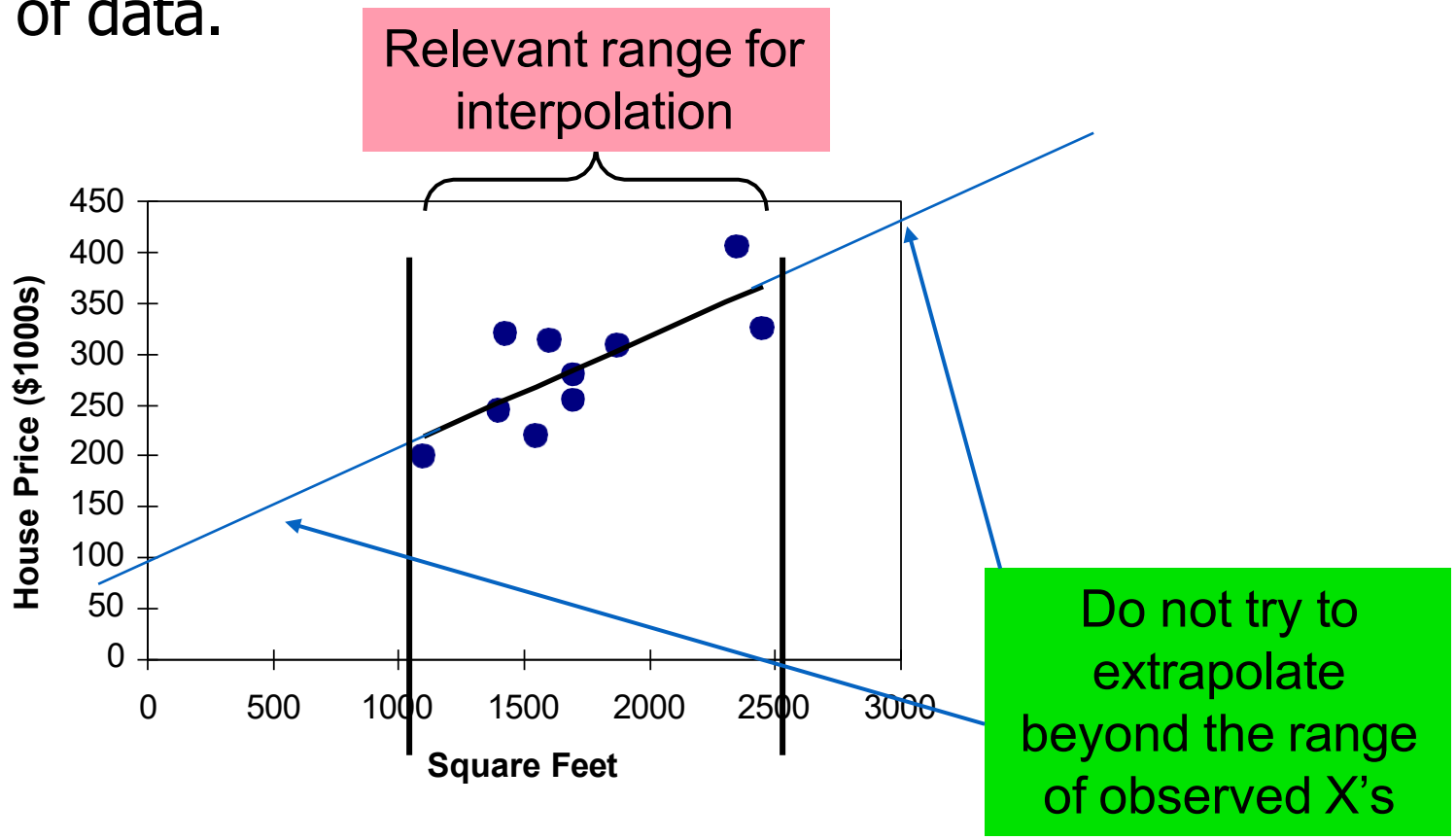
$$= 98.25 + 0.1098(2{,}000)$$

$$= 317.85$$

The predicted price for a house with 2,000 square feet is 317.85 ($1,000s) = $317,850

Simple Linear Regression Example: Making Predictions

When using a regression model for prediction, only predict within the relevant range of data.



Measures of Variation

Total variation is made up of two parts:

$$SST = SSR + SSE$$

Total Sum of Squares	Regression Sum of Squares	Error Sum of Squares
$SST = \sum (Y_i - \bar{Y})^2$	$SSR = \sum (\hat{Y}_i - \bar{Y})^2$	$SSE = \sum (Y_i - \hat{Y}_i)^2$

where:

$\bar{Y}$ = Mean value of the dependent variable

Y_i = Observed value of the dependent variable

$\hat{Y}_i$ = Predicted value of Y for the given X_i value

Measures of Variation

Total variation is made up of two parts:

SST = total sum of squares (Total Variation.)

- Measures the variation of the Y_i values around their mean Y.

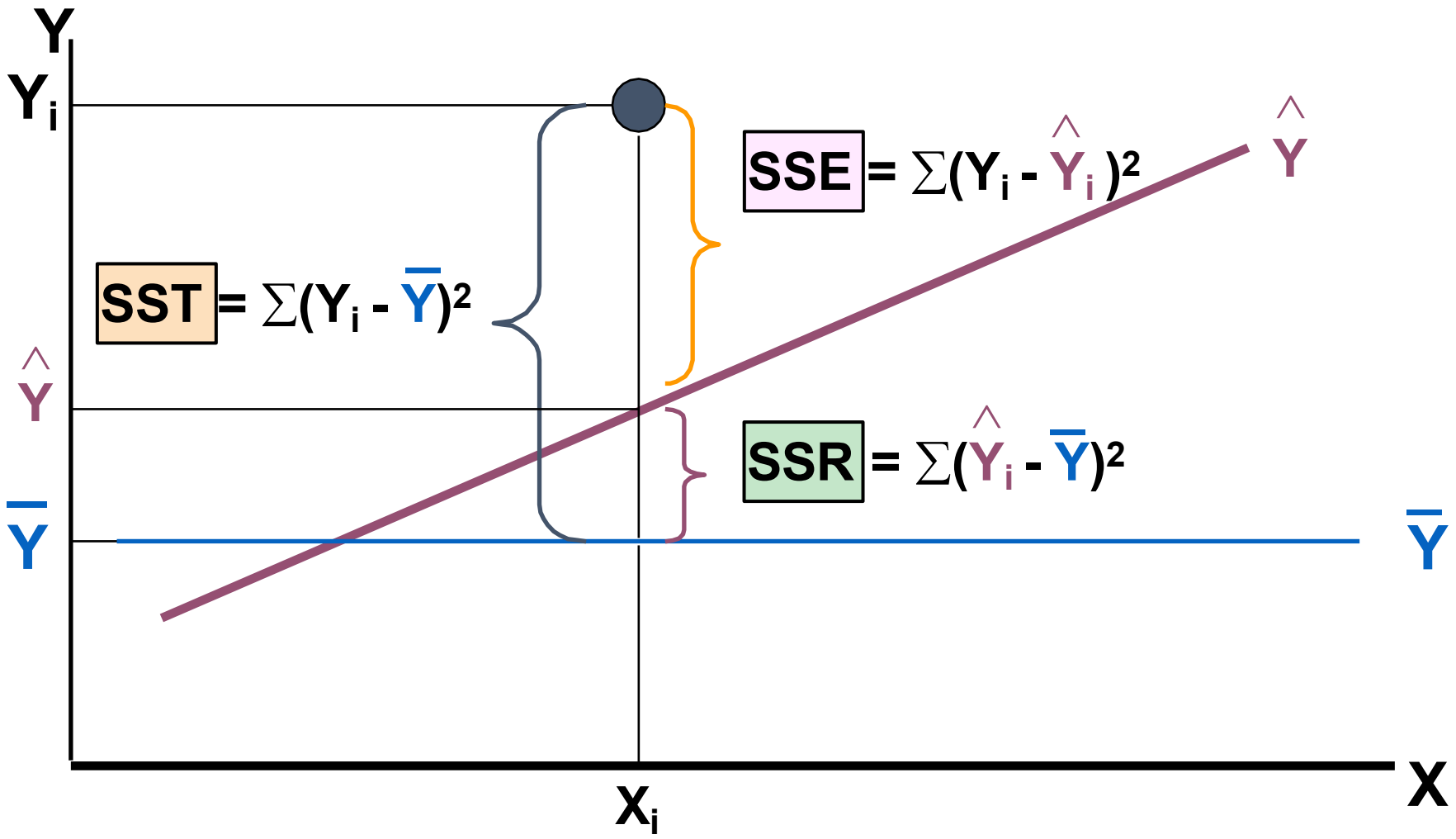
SSR = regression sum of squares (Explained Variation.)

- Variation attributable to the relationship between X and Y.

SSE = error sum of squares (Unexplained Variation.)

- Variation in Y attributable to factors other than X.

Measures of Variation



Excel Output of the Measures of Variation

10	ANOVA					
11		*df*	*SS*	*MS*	*F*	*Significance F*
12	Regression	1	18934.9348	18934.9348	11.0848	0.0104
13	Residual	8	13665.5652	1708.1957		
14	Total	9	32600.5000			

SST = SSR + SSE

32,600.5000 = 18,934.9348 + 13,665.5652

Coefficient of Determination, r^2

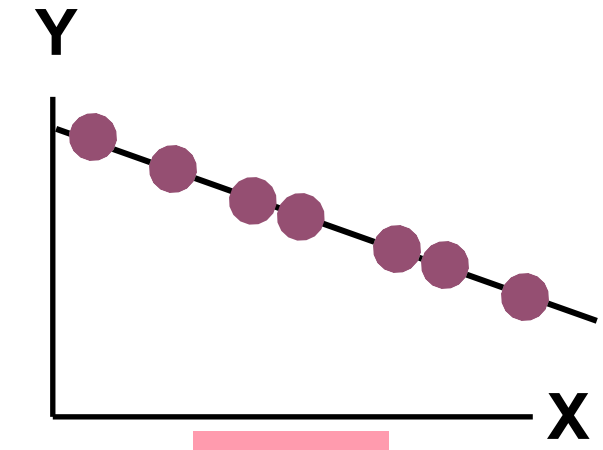
The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

The coefficient of determination is also called r-square and is denoted as r^2.

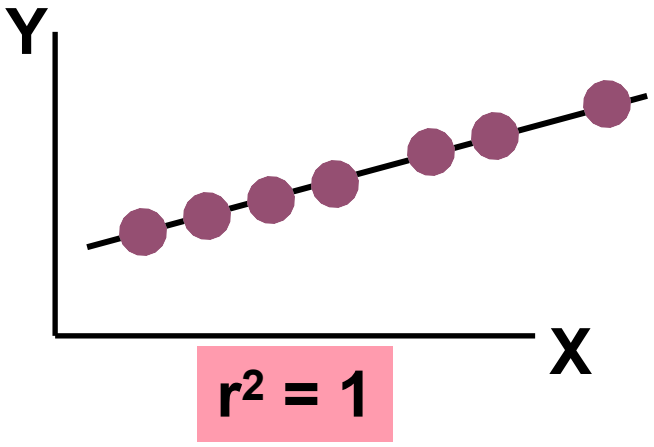
$$r^2 = \frac{SSR}{SST} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

$$0 \leq r^2 \leq 1$$

Examples of Approximate r^2 Values



$r^2 = 1$

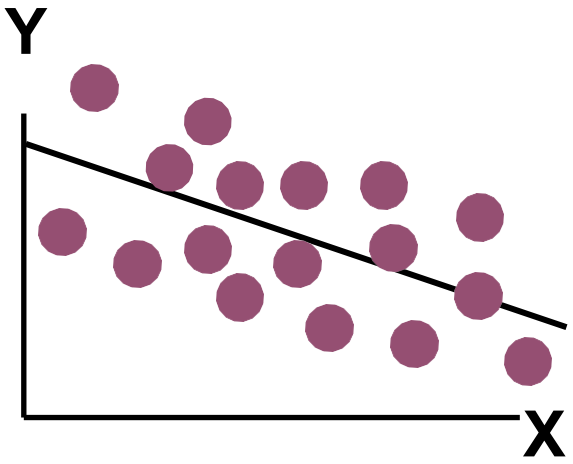


$r^2 = 1$

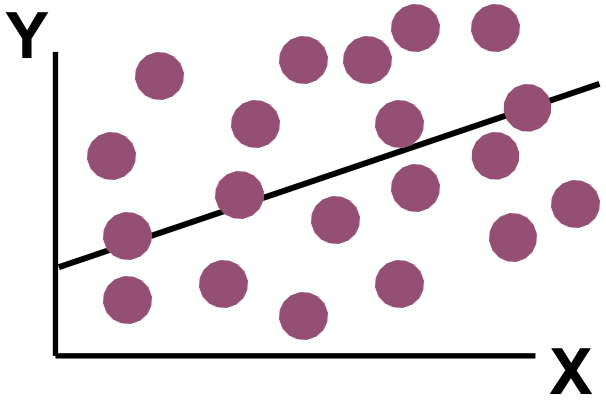
Perfect linear relationship between X and Y.

100% of the variation in Y is explained by variation in X.

Examples of Approximate r^2 Values



Weaker linear relationships between X and Y.



Some but not all of the variation in Y is explained by variation in X.

Simple Linear Regression Example: Coefficient of Determination, r^2 in Excel

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$r^2 = \frac{SSR}{SST} = \frac{18{,}934.9348}{32{,}600.5000} = 0.58082$$

58.08% of the variation in house prices is explained by variation in square feet.

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Standard Error of Estimate

The standard deviation of the variation of observations around the regression line is estimated by:

$$S_{YX} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}}$$

Where

SSE = error sum of squares

n = sample size

Standard Error of Estimate in Excel

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

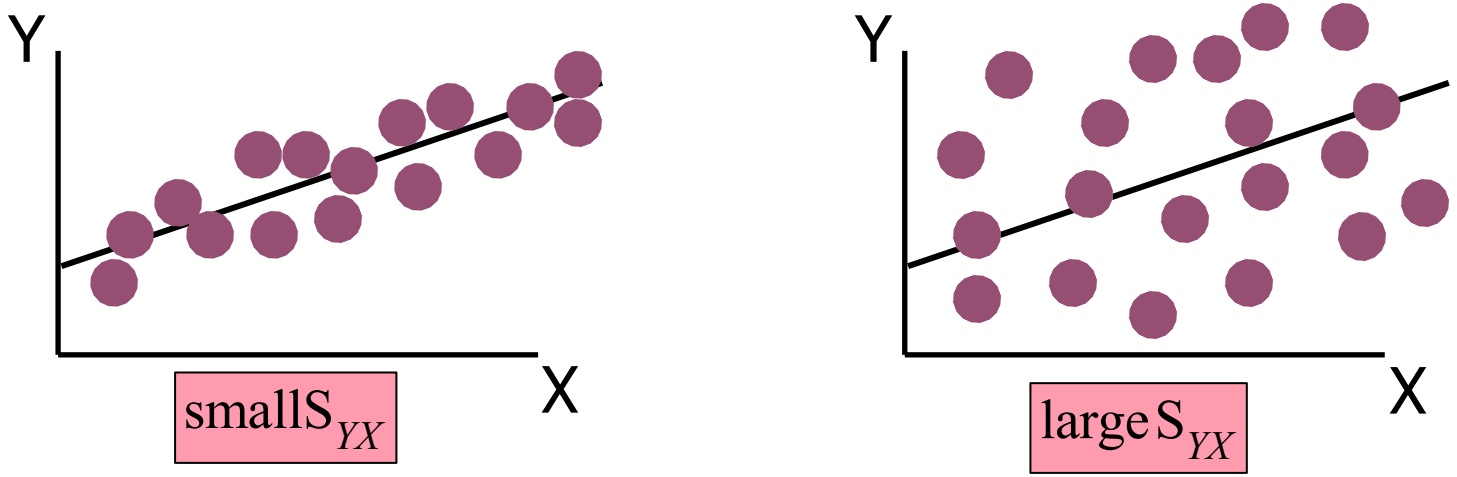
$S_{YX} = 41.33032$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Standard Error of Estimate in Excel

S_{YX} is a measure of the variation of observed Y values from the regression line.



The magnitude of S_{YX} should always be judged relative to the size of the Y values in the sample data.

i.e., S_{YX} = \$41.33K is moderately small relative to house prices in the \$200K - \$400K range

Assumptions of Regression (LINE)

- Linearity
  - The relationship between X and Y is linear
- Independence of Errors
  - Error values are statistically independent
- Normality of Error
  - Error values are normally distributed for any given value of X
- Equal Variance (also called homoscedasticity)
  - The probability distribution of the errors has constant variance

Assumptions of Regression (LINE)

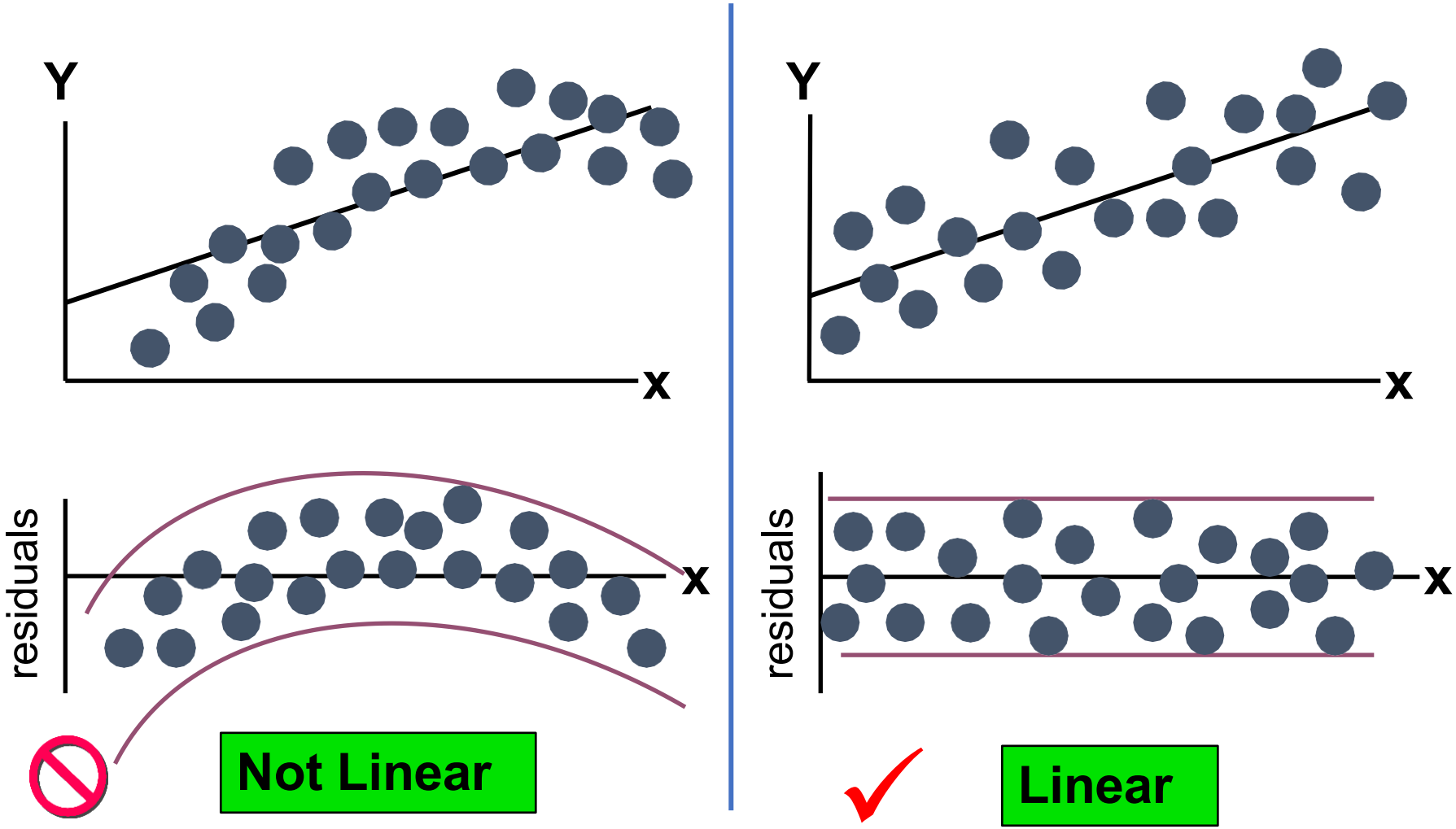
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  - Error values are statistically independent
- Normality of Error
  - Error values are normally distributed for any given value of X
- Equal Variance (also called homoscedasticity)
  - The probability distribution of the errors has constant variance

Residual Analysis

$$e_i = Y_i - \hat{Y}_i$$

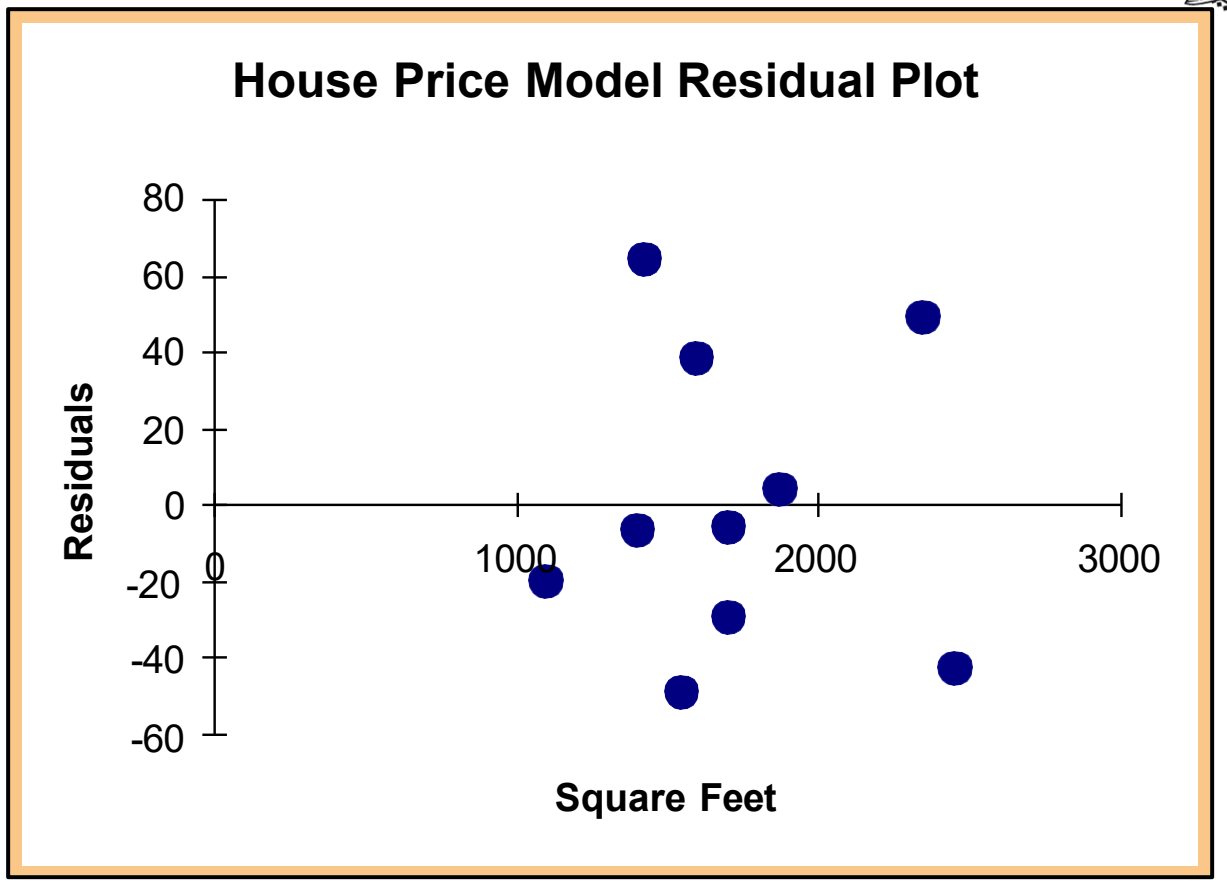
- The residual for observation i, e_i, is the difference between its observed and predicted value.
- Check the assumptions of regression by examining the residuals:
  - Examine for linearity assumption.
  - Evaluate independence assumption.
  - Evaluate normal distribution assumption.
  - Examine for constant variance (homoscedasticity) for all levels of X.
- Graphical Analysis of Residuals
  - Can plot residuals vs. X.

Residual Analysis for Linearity



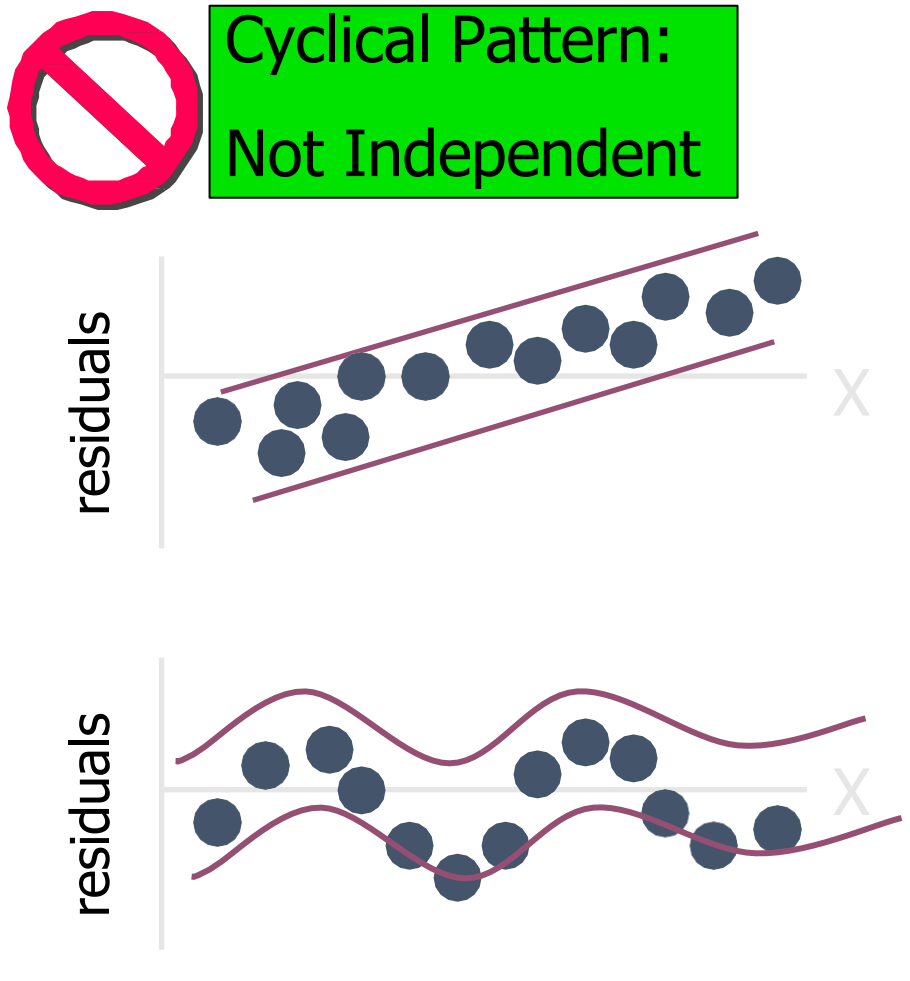
Residual Analysis: Checking for Linearity

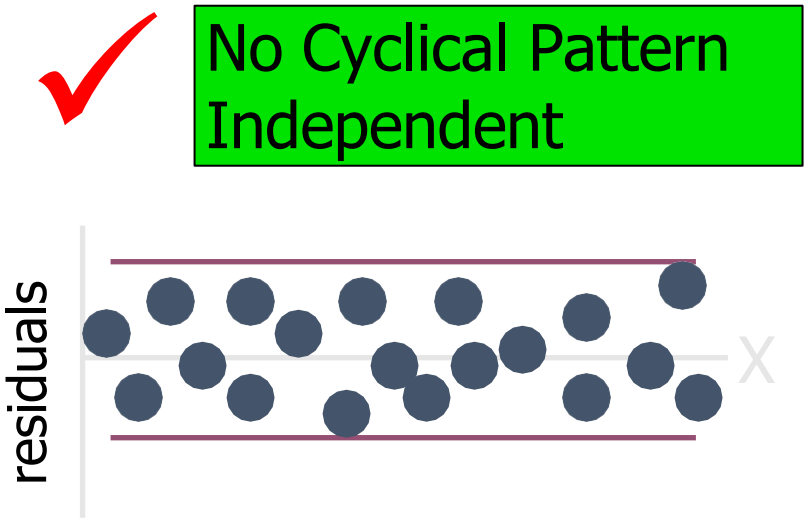
RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



Linear Model Assumption Is Appropriate

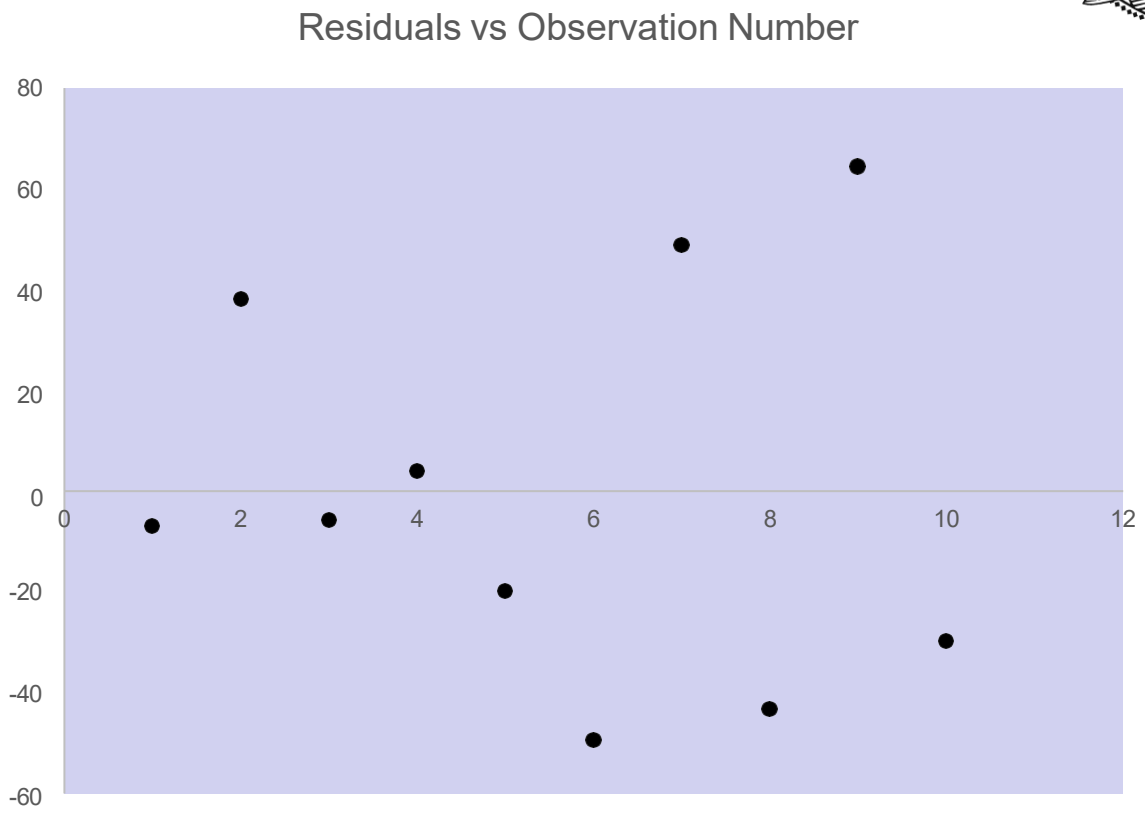
Residual Analysis for Independence





Residual Analysis: : Checking for Independence

RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
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9	254.6674	64.3326
10	284.8535	-29.8535



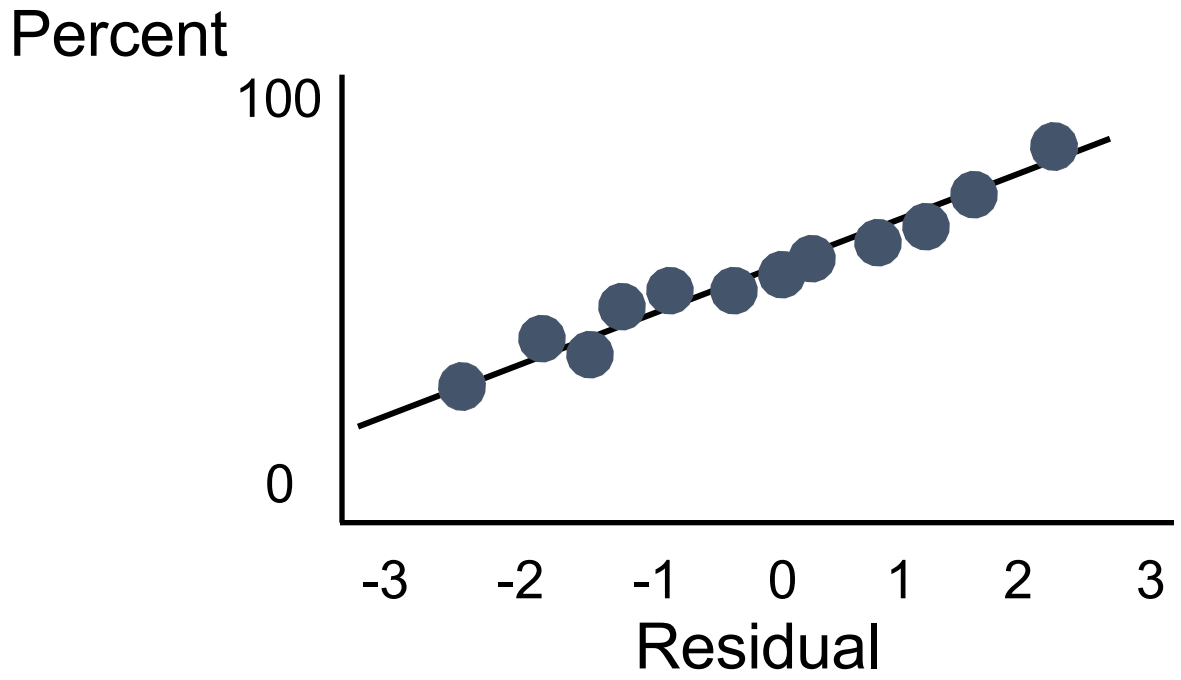
Independence Assumption Is Appropriate

Checking for Normality

- Examine the Stem-and-Leaf Display of the Residuals.
- Examine the Boxplot of the Residuals.
- Examine the Histogram of the Residuals.
- Construct a Normal Probability Plot of the Residuals.

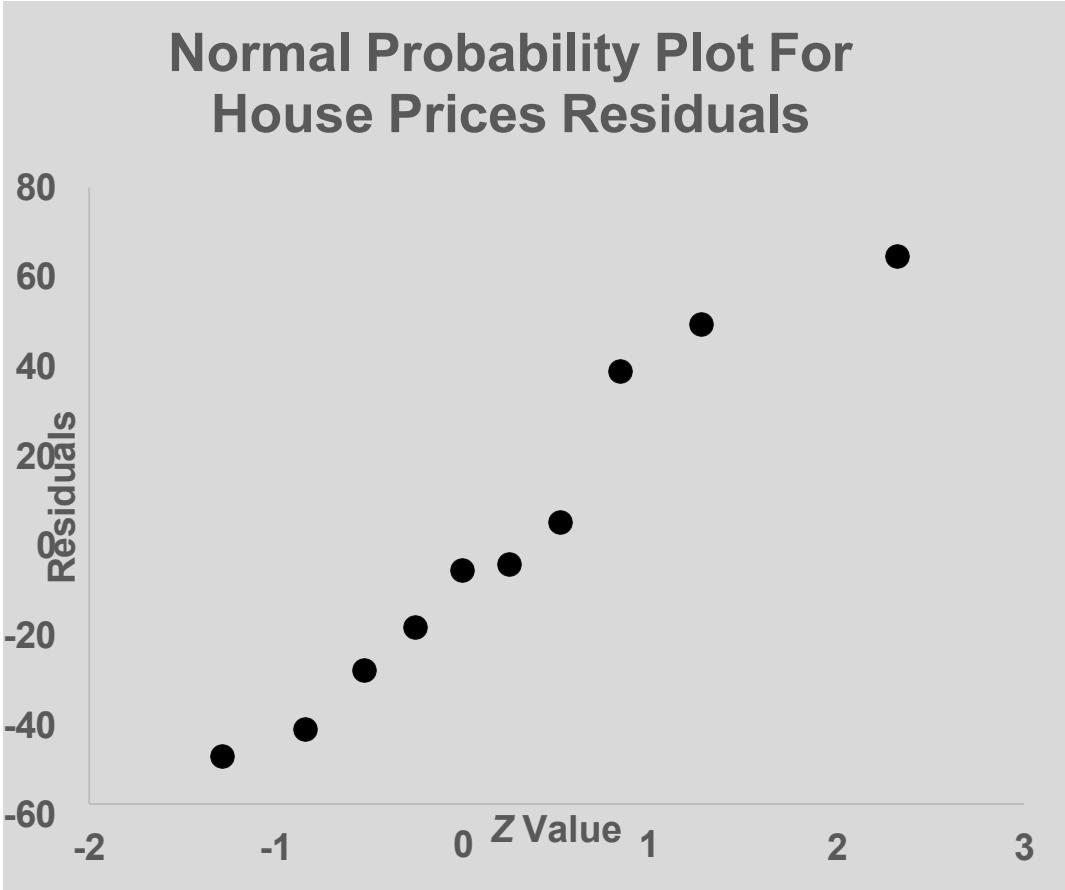
Residual Analysis for Normality

When using a normal probability plot, normal errors will approximately display in a straight line



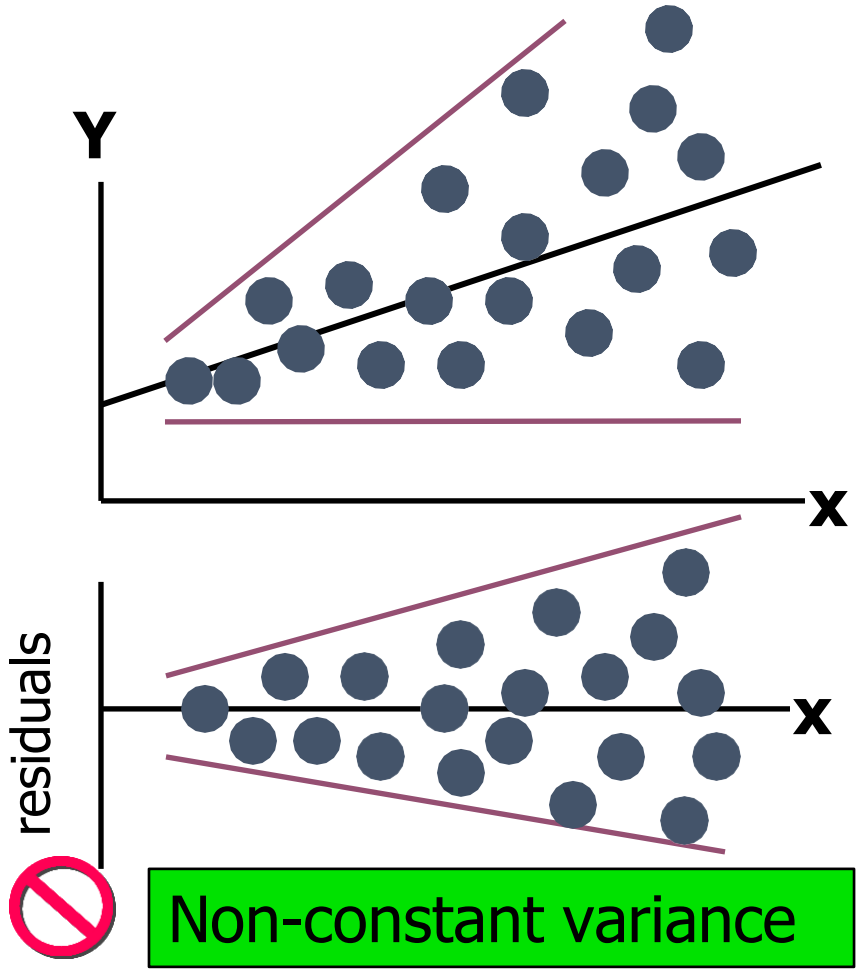
Residual Analysis: Checking for Normality

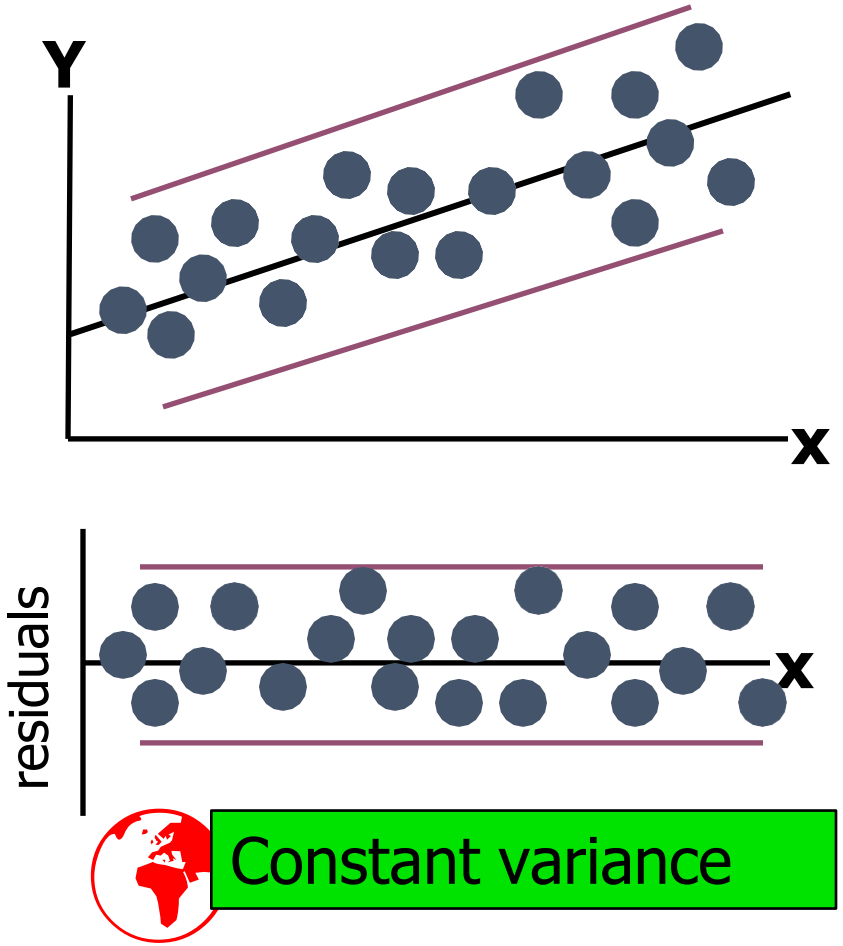
RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



Normality Assumption Is Appropriate

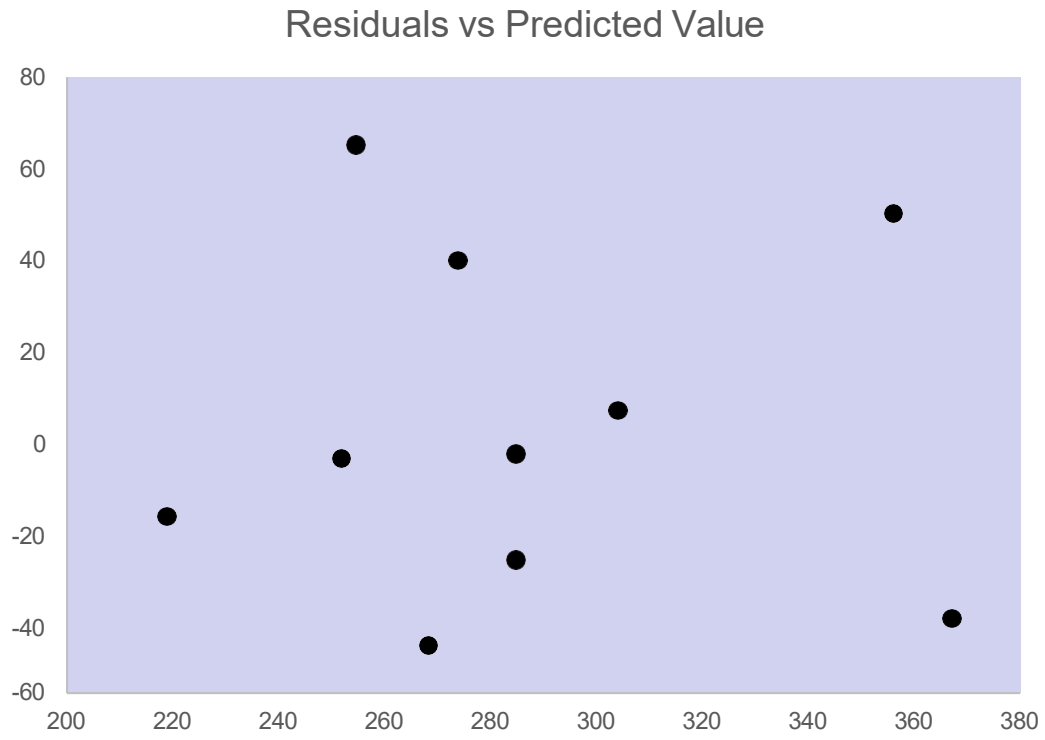
Residual Analysis for Equal Variance





Residual Analysis: Checking for Equal Variance

RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



Constant Variance Assumption Is Appropriate

Measuring Autocorrelation: The Durbin-Watson Statistic

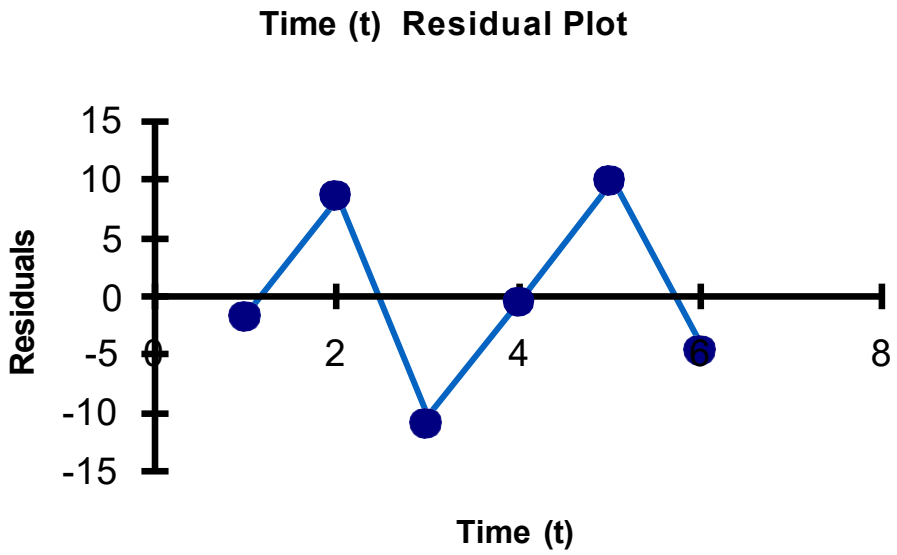
Used when data are collected over time to detect if autocorrelation is present.

Autocorrelation exists if residuals in one time period are related to residuals in another period.

Autocorrelation

Autocorrelation is correlation of the errors (residuals) over time.

- Here, residuals show a cyclical pattern, not random. Cyclical patterns are a sign of positive autocorrelation.



- Violates the regression assumption that residuals are random and independent.

The Durbin-Watson Statistic

Used to test for autocorrelation

H_0: positive autocorrelation does not exist

H_1: positive autocorrelation is present

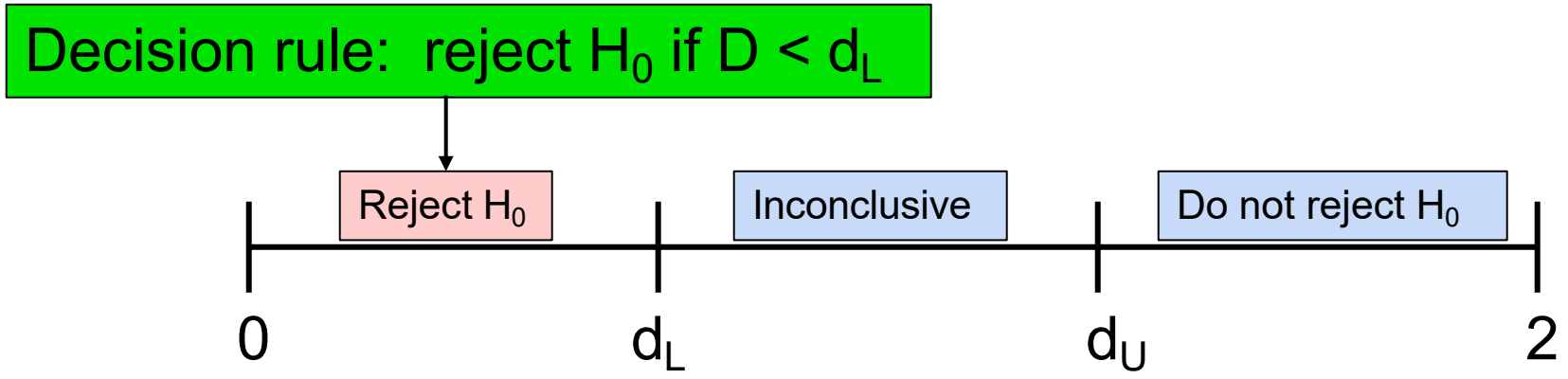
$$D = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

- The possible range is $0 \leq D \leq 4$.
- D should be close to 2 if H_0 is true.
- D less than 2 may signal positive autocorrelation, D greater than 2 may signal negative autocorrelation.

Testing for Positive Autocorrelation

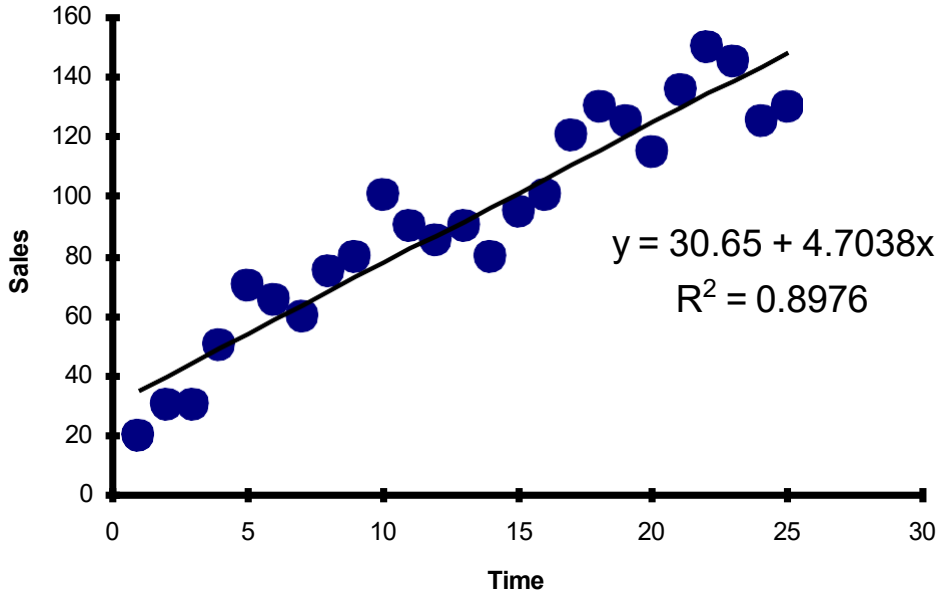
H_0: positive autocorrelation does not exist

H_1: positive autocorrelation is present



Testing for Positive Autocorrelation

- Suppose we have the following time series data:



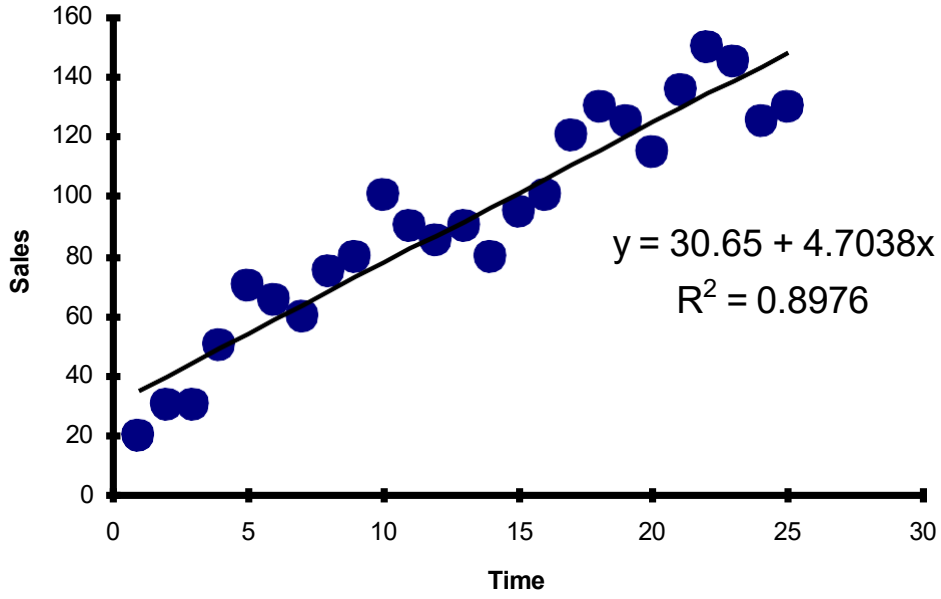
- Is there autocorrelation?

Testing for Positive Autocorrelation

- Example with n = 25:

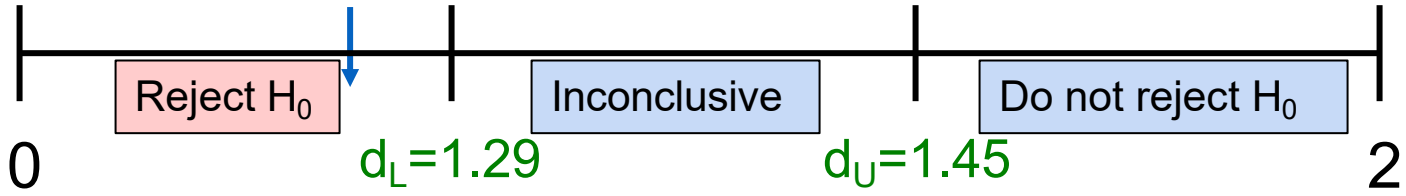
Excel/PHStat output:

Durbin-Watson Calculations	
Sum of Squared Difference of Residuals	3296.18
Sum of Squared Residuals	3279.98
Durbin-Watson Statistic	**1.00494**



$$D = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2} = \frac{3296.18}{3279.98} = 1.00494$$

Decision: reject H_0 since $D = 1.00494 < d_L$



Inferences about the Slope

The standard error of the regression slope coefficient (b_1) is estimated by:

$$S_{b_1} = \frac{S_{YX}}{\sqrt{SSX}} = \frac{S_{YX}}{\sqrt{\sum (X_i - \bar{X})^2}}$$

where:

S_{b_1} = Estimate of the standard error of the slope.

$S_{YX} = \sqrt{\frac{SSE}{n-2}}$ = Standard error of the estimate.

Inferences about the Slope: t Test

- t test for a population slope:
  - Is there a linear relationship between X and Y?
- Null and alternative hypotheses:
  - H_0: $\beta_1 = 0$ (no linear relationship)
  - H_1: $\beta_1 \neq 0$ (linear relationship does exist)
- Test statistic :

$$t_{STAT} = \frac{b_1 - \beta_1}{S_{b_1}}$$

$$d.f. = n - 2$$

where:

b_1 = regression slope coefficient

β_1 = hypothesized slope

S_{b1} = standard error of the slope

Inferences about the Slope: t Test

House Price in $1000s (y)	Square Feet (x)
245	1400
312	1600
279	1700
308	1875
199	1100
219	1550
405	2350
324	2450
319	1425
255	1700

Estimated Regression Equation:

$$\text{house price} = 98.25 + 0.1098\ (\text{sq.ft.})$$

The slope of this model is 0.1098

Is there a relationship between the square footage of the house and its sales price?

Inferences about the Slope: t Test

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

From Excel output:

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	98.24833	58.03348	1.69296	0.12892
Square Feet	0.10977	0.03297	3.32938	0.01039

b_1 S_{b_1}

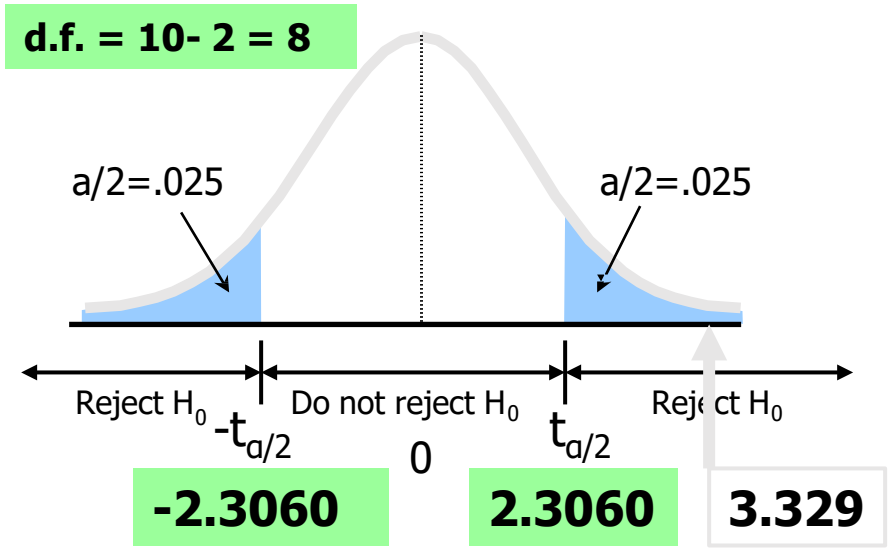
$$t_{STAT} = \frac{b_1 - \beta_1}{S_{b_1}} = \frac{0.10977 - 0}{0.03297} = 3.32938$$

Inferences about the Slope: t Test

Test Statistic: $t_{STAT} = 3.329$

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$



Decision: Reject H_0.

There is sufficient evidence that square footage affects house price.

Inferences about the Slope: t Test

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

From Excel output:

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	98.24833	58.03348	1.69296	0.12892
Square Feet	0.10977	0.03297	3.32938	0.01039

p-value

Decision: Reject H_0, since p-value < α.

There is sufficient evidence that square footage affects house price.

Confidence Interval Estimate for the Slope

$$b_1 \pm t_{\alpha/2} S_{b_1}$$

d.f. = n - 2

Excel Printout for House Prices:

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

At 95% level of confidence, the confidence interval for the slope is (0.0337, 0.1858).

Confidence Interval Estimate for the Slope

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Since the units of the house price variable is $1,000s, we are 95% confident that the average impact on sales price is between $33.74 and $185.80 per square foot of house size

This 95% confidence interval does not include 0.

Conclusion: There is a significant relationship between house price and square feet at the .05 level of significance.

Finding Confidence and Prediction Intervals in Excel



- From Excel, use:

  PHStat | regression | simple linear regression …

  - Check the
    "confidence and prediction interval for X="
    box and enter the X-value and confidence level desired.

Finding Confidence and Prediction Intervals in Excel

	A	B
1	**Confidence Interval Estimate**	
2		
3	**Data**	
4	**X Value**	**2000**
5	**Confidence Level**	**95%**
6		
7	Intermediate Calculations	
8	Sample Size	10
9	Degrees of Freedom	8
10	t Value	2.306006
11	Sample Mean	1715
12	Sum of Squared Difference	1571500
13	Standard Error of the Estimate	41.33032
14	h Statistic	0.151686
15	Average Predicted Y (YHat)	317.7838
16		
17	**For Average Predicted Y (YHat)**	
18	Interval Half Width	37.11952
19	**Confidence Interval Lower Limit**	**280.6643**
20	**Confidence Interval Upper Limit**	**354.9033**
21		
22	**For Individual Response Y**	
23	Interval Half Width	102.2813
24	**Prediction Interval Lower Limit**	**215.5025**
25	**Prediction Interval Upper Limit**	**420.0651**

Input values

$\hat{Y}$

Confidence Interval Estimate for $\mu_{Y|X=Xi}$

Prediction Interval Estimate for $Y_{X=Xi}$

Pitfalls of Regression Analysis

- Lacking an awareness of the assumptions underlying least-squares regression
- Not knowing how to evaluate the assumptions
- Not knowing the alternatives to least-squares regression if a particular assumption is violated
- Using a regression model without knowledge of the subject matter
- Extrapolating outside the relevant range

Strategies for Avoiding Pitfalls of Regression

- Start with a scatter plot of X vs. Y to observe possible relationship
- Perform residual analysis to check the assumptions
  - Plot the residuals vs. X to check for violations of assumptions such as homoscedasticity
  - Use a histogram, stem-and-leaf display, boxplot, or normal probability plot of the residuals to uncover possible non-normality

Strategies for Avoiding Pitfalls of Regression

- If there is violation of any assumption, use alternative methods or models.
- If there is no evidence of assumption violation, then test for the significance of the regression coefficients and construct confidence intervals and prediction intervals.
- Avoid making predictions or forecasts outside the relevant range.

Part of Normal probability Distribution
Sampling Distribution
Central Limit theorem
Confidence interval
Hypothesis testing
Simple linear regression
Multiple linear regression
Hypothesis testing in two sample test

END OF CHAPTER 13